



(12) **United States Plant Patent**  
**Gilford et al.**

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(54) **STRAWBERRY PLANT NAMED ‘DRISCOLL ATLANTIS’**

(50) Latin Name: *Fragaria*×*ananassa*  
Varietal Denomination: **Driscoll Atlantis**

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See application file for complete search history.

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(57) **ABSTRACT**

This invention relates to a new and distinct variety of  
strawberry named ‘Driscoll Atlantis.’ The variety is similar  
to the varieties ‘Biscayne’ and ‘Key Largo.’ The variety is  
distinguished from ‘Biscayne’ and ‘Key Largo,’ in  
particular, by its shorter average stipule length, reduced  
frequency of paired bracts, shorter sepal length and width,  
weak leaf glossiness, dense stipule pubescence, and a very  
narrow band on the fruit where achenes are absent.

**5 Drawing Sheets**

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1. Latin name of the genus and species of the plant  
claimed: The variety is botanically identified as *Fragaria*×  
*ananassa*.

1.1 Variety denomination: The strawberry variety  
denomination is ‘Driscoll Atlantis’.

**2. BACKGROUND OF THE INVENTION**

The new variety originated as a result of a controlled cross  
between the strawberry plants ‘88E94’ (unpatented variety)  
and Mirador (U.S. Plant Pat. No. : 11,279) in an ongoing  
breeding program, and was discovered as a seedling in  
Hillsborough, Fla. in 1999. The original seedling of the new  
cultivar was asexually propagated by stolons in a Nursery in  
Shasta County, Calif. Propagules were transplanted to a  
controlled breeding plot in Hillsborough County, Fla., where  
the variety was identified and selected for further evaluation.  
‘Driscoll Atlantis’ was subsequently asexually propagated  
and underwent further testing in Hillsborough County, Fla.  
for three years. This propagation and testing has demon-  
strated that the combination of traits disclosed herein which  
characterize the new variety are fixed and retained true to  
type through successive generations of asexual reproduc-  
tion.

**3. SUMMARY OF THE INVENTION**

The present invention relates to a new and distinct variety  
of strawberry named ‘Driscoll Atlantis.’ The variety is  
botanically identified as *Fragaria*×*ananassa*. The new vari-  
ety is distinguished from other varieties by a number of  
characteristics as set forth in Tables 1–4.

**4. COMPARISON TO SIMILAR VARIETIES**

The varieties which we believe to be similar to ‘Driscoll  
Atlantis’ from those known to us are ‘Biscayne’ (U.S. Plant  
Pat. No. 12,186) and ‘Key Largo’ ( U.S. Plant Pat. No.  
8,649). There are several characteristics of the new variety  
that are different from, or not possessed by ‘Biscayne’ and  
‘Key Largo’. The new variety has a shorter average stipule

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length, a reduced frequency of paired bracts, a shorter sepal  
length and width, weak leaf glossiness, dense stipule  
pubescence, and a very narrow band on the fruit where  
achenes are absent.

5 ‘Driscoll Atlantis’ is distinguished from its parents  
‘88E94’ and ‘Mirador’ by being partially everbearing, in  
comparison to the fully everbearing trait of both parents.

**5. BRIEF DESCRIPTION OF THE DRAWINGS**

10 The accompanying photographs show typical specimens  
of the new variety, including fruit, foliage and flowers, in  
color as nearly true as it is reasonably possible to make in  
color illustrations of these characteristics.

15 FIG. 1 shows the upper side of the leaves of the plant.

FIG. 2 shows the whole the plant.

FIG. 3 shows the under side and underside of the flowers.

FIG. 4 shows a close-up of the fruit.

20 FIG. 5 shows the fruit in longitudinal cross-section.

**6. DESCRIPTION OF THE NEW VARIETY**

25 The following detailed description of the new variety is  
based upon observations taken of plants and fruit grown in  
Hillsborough County, Fla., U.S.A. This description is in  
accordance with UPOV terminology. Observations of  
‘Driscoll Atlantis’, ‘Biscayne’ and ‘Key Largo’ were taken  
in side by side comparison in the 2003–2004 winter season.  
Plants for observation were harvested from McArthur,  
Calif., and held in refrigerated storage until planting in  
Hillsborough County, Fla. in October 2003. Plants were  
grown in raised beds of soil under conditions typical of  
commercial strawberry production in central Florida. Fruits  
were harvested twice weekly for yield determination from  
November 2003 to March 2004. Measurements of plant,  
flower, and fruit characteristics were made in January 2004,  
approximately four months after planting. Color  
designations, color descriptions, and other phenotypical  
descriptions may deviate from the stated values and descrip-



tions depending upon variation in environmental, seasonal, climatic and cultural conditions. Colors are described and the most similar color designations are provided from The Royal Horticultural Society (R.H.S.) Colour Chart.

6.1 PROPAGATION

The new variety is principally propagated by way of stolons. Although propagation by stolons is presently preferred, other known methods of propagating strawberry plants may be employed.

6.2 CHARACTERISTICS OF THE NEW VARIETY

Information on the new variety is presented in Tables 1, 2, 3 and 4. In the tables, the flowers described are secondary flowers except where indicated. The fruit described is the secondary fruit on five month old plants. Fruit and flower measurements are an average of both primary and secondary fruit and flowers.

Table 1 provides information on the plant and fruit characteristics of the new variety ‘Driscoll Atlantis’ compared with characteristics of ‘Biscayne’ and ‘Key Largo.’ Table 2 provides additional information of the plant and fruit characteristics of the new variety ‘Driscoll Atlantis’ compared with characteristics of the varieties ‘Biscayne’ and ‘Key Largo.’ Table 3 provides reactions of the new variety to stresses, pests and diseases compared with reactions of the varieties ‘Biscayne’ and ‘Key Largo.’ Table 4 provides isozyme characteristics of the new variety as compared to that of the varieties ‘Biscayne’ and ‘Key Largo.’

TABLE 1

| QUANTITATIVE COMPARISON OF ‘DRISCOLL ATLANTIS’, ‘BISCAYNE,’ AND ‘KEY LARGO’ |                       |                       |                       |
|-----------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|
|                                                                             | ‘Driscoll Atlantis’   | ‘Biscayne’            | ‘Key Largo’           |
| Plant Characteristics                                                       |                       |                       |                       |
| Height of Plant (cm)                                                        | 8.7                   | 10.0                  | 11.6                  |
| Spread of Plant (cm)                                                        | 30.8                  | 31.3                  | 32.1                  |
| Number of Crowns                                                            | 3.7                   | 3.2                   | 4.2                   |
| Leaf Characteristics                                                        |                       |                       |                       |
| Terminal Leaflet Length (cm)                                                | 7.0                   | 7.6                   | 8.4                   |
| Terminal Leaflet Width (cm)                                                 | 7.2                   | 8.1                   | 8.0                   |
| Terminal Leaflet Length/Width Ratio                                         | 0.97                  | 0.93                  | 1.06                  |
| # Teeth/Terminal Leaflet                                                    | 19.0                  | 27.0                  | 21.6                  |
| Color of upper side                                                         | 137A medium green     | 137A light green      | 137A light green      |
| Color of under side                                                         | 140D light gray green | 139C light gray green | 139C light gray green |
| Petiole Length (cm)                                                         | 10.5                  | 9.2                   | 11.5                  |
| Petiole Color                                                               | 145C yellow green     | 142D yellow green     | 142D yellow green     |
| Petiolule Length (mm)                                                       | 4.7                   | 5.5                   | 6.3                   |
| Petiolule Diameter (mm)                                                     | 1.7                   | 2.2                   | 1.9                   |
| Petiolule Color                                                             | 145C yellow green     | 142D yellow green     | 142D yellow green     |
| Bract Frequency                                                             | 50% typically paired  | 83% typically paired  | 0%                    |
| Stipule Length (cm)                                                         | 2.9                   | 3.7                   | 3.4                   |
| Stipule Width (cm)                                                          | 1.5                   | 2.0                   | 1.5                   |

TABLE 1-continued

| QUANTITATIVE COMPARISON OF ‘DRISCOLL ATLANTIS’, ‘BISCAYNE,’ AND ‘KEY LARGO’ |                                       |                                       |                                       |
|-----------------------------------------------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
|                                                                             | ‘Driscoll Atlantis’                   | ‘Biscayne’                            | ‘Key Largo’                           |
| Stolon Characteristics                                                      |                                       |                                       |                                       |
| Anthocyanin color                                                           | 60A red                               | 59B red                               |                                       |
| Diameter at bract (mm)                                                      | 2.92                                  | 3.24                                  | 3.30                                  |
| Avg. # of Daughter plants (2003 Nursery)                                    | 53                                    | 72                                    | 52                                    |
| Flower Characteristics                                                      |                                       |                                       |                                       |
| Petal Length (cm)                                                           | 1.3                                   | 1.36                                  | 1.65                                  |
| Petal Width (cm)                                                            | 1.3                                   | 1.49                                  | 1.46                                  |
| Petal Length/Width Ratio                                                    | 0.96                                  | 0.91                                  | 1.13                                  |
| Flower Diameter (cm)                                                        | 2.85                                  | 2.76                                  | 2.72                                  |
| Calyx Diameter (cm)                                                         | 3.48                                  | 3.86                                  | 5.11                                  |
| Sepal Length (mm)                                                           | 12.7                                  | 15.3                                  | 21.5                                  |
| Sepal Width (cm)                                                            | 6.8                                   | 7.22                                  | 9.15                                  |
| Petal Color                                                                 | 155C white                            | 155C white                            | 155C white                            |
| Receptical Color                                                            | 149C yellow green                     | 150A yellow green                     | 150A yellow green                     |
| Anther Color                                                                | 9A yellow                             | 9A yellow                             | 9A yellow                             |
| Fruiting Truss Length (cm)                                                  | 18.2                                  | 8.7                                   | 13.6                                  |
| Fruit Characteristics                                                       |                                       |                                       |                                       |
| Fruit Length (cm)                                                           | 5.0                                   | 4.8                                   | 4.8                                   |
| Fruit Width (cm)                                                            | 3.9                                   | 4.0                                   | 3.5                                   |
| Fruit Length/Width Ratio                                                    | 1.29                                  | 1.20                                  | 1.37                                  |
| Average Berry Weight (g)                                                    | 23.6                                  | 26.5                                  | 23.2                                  |
| External Color                                                              | 45B red                               | 46B red                               | 46B red                               |
| Internal Color                                                              | 42B dark red                          | 44A medium red                        | 44B medium red                        |
| Achene Coloration                                                           | 151D to 185D grey red to yellow green | 180B to 150C grey red to yellow green | 182B to 150A grey red to yellow green |
| Achenes per berry                                                           | 121                                   | 109                                   | 104                                   |
| Achene weight (g)                                                           | 0.000082                              | 0.00046                               | 0.00058                               |
| 2003–2004 Fruit Yield (g/plant)                                             | 327                                   | 243                                   | 251                                   |

TABLE 2

| QUALITATIVE COMPARISON OF ‘DRISCOLL ATLANTIS’, ‘BISCAYNE,’ AND ‘KEY LARGO’ |                          |              |                  |
|----------------------------------------------------------------------------|--------------------------|--------------|------------------|
|                                                                            | ‘Driscoll Atlantis’      | ‘Biscayne’   | ‘Key Largo’      |
| Plant                                                                      |                          |              |                  |
| Habit                                                                      | flat globose             | flat globose | globose          |
| Canopy Density                                                             | medium to dense          | medium       | medium           |
| Vigor                                                                      | medium                   | strong       | medium           |
| Leaf                                                                       |                          |              |                  |
| Shape in cross section                                                     | slightly concave to flat | concave      | slightly concave |
| Interveinal blistering                                                     | medium                   | weak         | weak             |
| Glossiness                                                                 | weak                     | medium       | medium           |
| Number of leaflets                                                         | three only               | three only   | three only       |
| Terminal leaflet margin profile                                            | flat                     | revolute     | revolute         |
| Terminal leaflet shape of base                                             | rounded                  | rounded      | slightly oblique |



TABLE 2-continued

| QUALITATIVE COMPARISON OF ‘DRISCOLL ATLANTIS’,<br>‘BISCAYNE,’ AND ‘KEY LARGO’ |                               |                                   |                               |
|-------------------------------------------------------------------------------|-------------------------------|-----------------------------------|-------------------------------|
|                                                                               | ‘Driscoll<br>Atlantis’        | ‘Biscayne’                        | ‘Key Largo’                   |
| Terminal leaflet shape<br>of teeth                                            | obtuse to<br>rounded          | rounded                           | rounded                       |
| Stipule pubescence                                                            | dense                         | medium                            | medium                        |
| Petiole pubescence                                                            | dense                         | medium                            | medium                        |
| Petiole pose of hairs                                                         | outwards                      | outwards                          | upward                        |
| <u>Stolon</u>                                                                 |                               |                                   |                               |
| Anthocyanin<br>coloration                                                     | weak to<br>medium             | strong                            |                               |
| Thickness                                                                     | medium                        | medium                            |                               |
| Pubescence                                                                    | medium                        | medium                            |                               |
| <u>Inflorescence</u>                                                          |                               |                                   |                               |
| Position relative to<br>foliage                                               | level to above                | level                             | level to above                |
| Diameter of calyx<br>relative to corolla on<br>secondary flowers              | same size to<br>larger        | larger                            | much larger                   |
| Diameter of inner<br>calyx relative to outer<br>on secondary flowers          | same size                     | same size                         | smaller                       |
| Spacing of petals                                                             | overlapping                   | overlapping                       | overlapping                   |
| <u>Fruiting Truss</u>                                                         |                               |                                   |                               |
| Attitude at first<br>picking                                                  | prostrate                     | prostrate                         | semi-erect                    |
| <u>Fruit</u>                                                                  |                               |                                   |                               |
| Predominant shape                                                             | conical                       | conical                           | conical                       |
| Difference in shapes<br>between primary and<br>secondary fruits               | slight                        | slight                            | slight                        |
| Band without achenes                                                          | very narrow                   | narrow                            | narrow                        |
| Unevenness of surface                                                         | weak to<br>medium             | weak                              | weak                          |
| Evenness of color                                                             | even                          | even                              | even                          |
| Glossiness                                                                    | strong                        | strong                            | strong                        |
| Insertion of achenes                                                          | level with<br>surface         | level with<br>surface             | below surface                 |
| Insertion of calyx                                                            | level                         | level                             | set above                     |
| Pose of the calyx<br>segments                                                 | spreading to<br>reflexed      | spreading                         | reflexed                      |
| Size of calyx in<br>relation to fruit on<br>secondary fruit                   | same to larger                | larger                            | larger                        |
| Adherence of calyx                                                            | strong                        | strong                            | strong                        |
| Firmness of flesh                                                             | firm                          | firm                              | firm                          |
| Evenness of flesh<br>color                                                    | slightly uneven               | slightly uneven                   | slightly uneven               |
| Distribution of flesh<br>color                                                | marginal and<br>central       | marginal and<br>central           | marginal and<br>central       |
| Hollow center size                                                            | medium                        | large                             | medium                        |
| Sweetness                                                                     | strong                        | strong                            | medium                        |
| Texture when tasted                                                           | medium                        | fine                              | medium                        |
| Acidity                                                                       | medium                        | medium                            | medium                        |
| Time of First<br>Flowering in 2003–<br>2004                                   | Late October                  | Mid-November                      | Late November                 |
| Harvest Interval in<br>2003–2004                                              | Late November<br>to Mid-March | Early<br>December to<br>Mid-March | Late December<br>to Mid-March |
| Type of Bearing                                                               | partially<br>everbearing      | partially<br>everbearing          | partially<br>everbearing      |

6.3 REACTION TO STRESS, PESTS, AND  
DISEASE

TABLE 3

| REACTIONS TO STRESS PESTS AND DISEASES FOR ‘DRISCOLL<br>ATLANTIS’, ‘BISCAYNE,’ AND ‘KEY LARGO’ |                           |                           |                           |
|------------------------------------------------------------------------------------------------|---------------------------|---------------------------|---------------------------|
|                                                                                                | ‘Driscoll<br>Atlantis’    | ‘Biscayne’                | ‘Key Largo’               |
| <u>Reaction to Pests</u>                                                                       |                           |                           |                           |
| <i>Tetranychus urticae</i>                                                                     | susceptible               | susceptible               | moderately<br>susceptible |
| <i>Lygus hesperus</i>                                                                          | susceptible               | susceptible               | susceptible               |
| <u>Reaction To Diseases</u>                                                                    |                           |                           |                           |
| Botrytis fruit rot                                                                             | susceptible               | susceptible               | susceptible               |
| Powdery mildew                                                                                 | susceptible               | moderately<br>susceptible | susceptible               |
| <i>Verticillium</i> wilt                                                                       | moderately<br>susceptible | susceptible               | susceptible               |
| Strawberry Mottle Virus                                                                        | moderately<br>resistant   | moderately<br>resistant   | moderately<br>resistant   |
| <i>Xanthomonas fragariae</i>                                                                   | moderately<br>susceptible | moderately<br>susceptible | moderately<br>susceptible |

6.4 ISOZYME ANALYSIS

In addition to the morphological description above, the new cultivar ‘Driscoll Atlantis’ has been analyzed to obtain an indication of its genetic makeup to provide further means for identifying the new variety and distinguishing it from other somewhat similar and/or related strawberry varieties. Specifically, leaf samples of ‘Driscoll Atlantis’, ‘Biscayne’ and ‘Key Largo’ were analyzed by electrophoresis for isozyme patterns of the enzymes phosphoglucosomerase (“PGI”), leucine aminopeptidase (“LAP”) and phosphoglucosomutase (“PGM”). See J. Amer. Soc. Hort. Sci. 106:684–687. Isozyme characterization of the three varieties is presented in Table 4, with the letters representing the banding patterns for each enzyme as designated in the above-identified article.

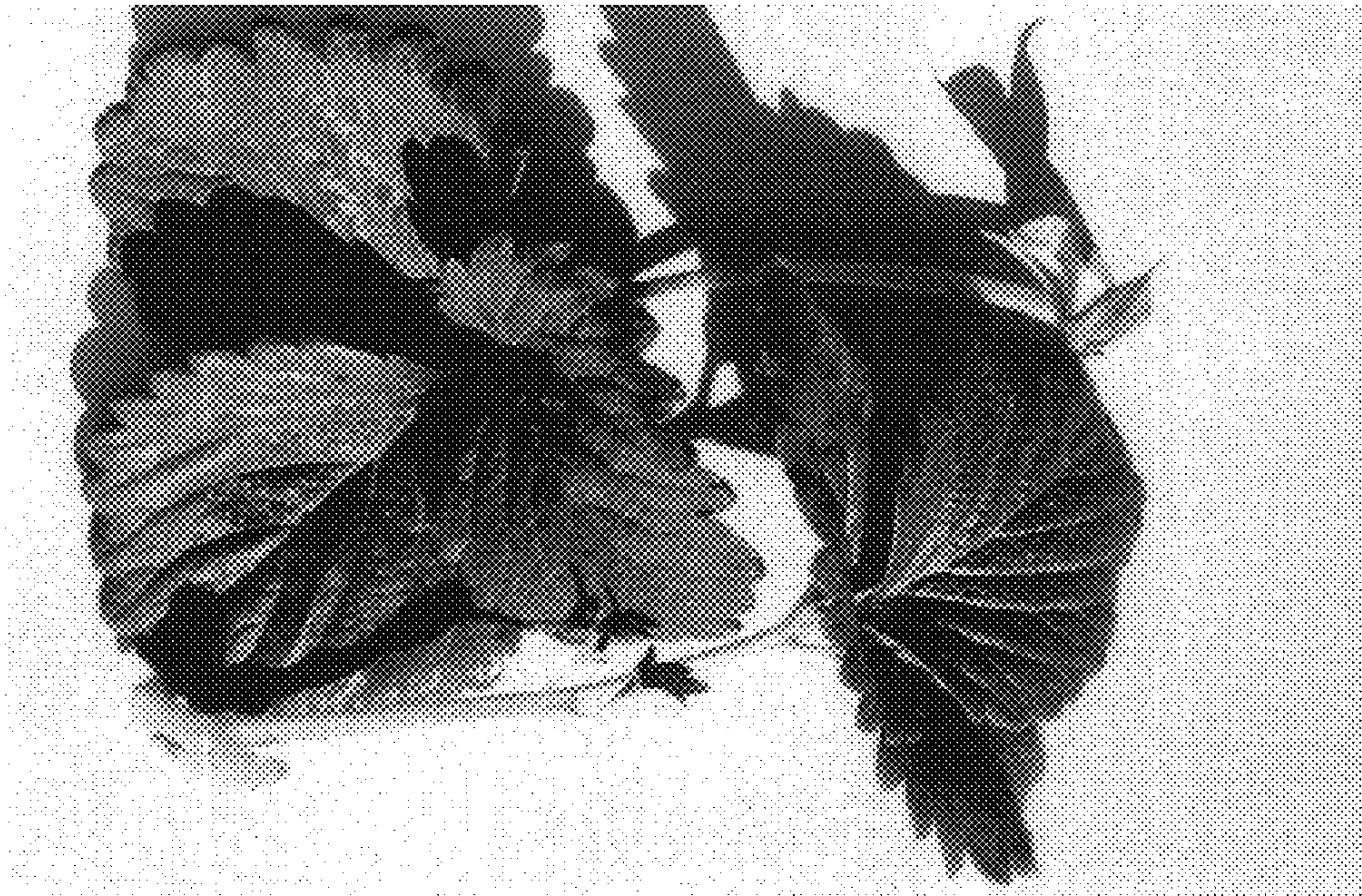
TABLE 4

| ISOZYME ANALYSIS FOR ‘DRISCOLL ATLANTIS’,<br>‘BISCAYNE,’ AND ‘KEY LARGO’ |                        |            |             |
|--------------------------------------------------------------------------|------------------------|------------|-------------|
| Locus                                                                    | ‘Driscoll<br>Atlantis’ | ‘Biscayne’ | ‘Key Largo’ |
| PGI                                                                      | A4                     | A1         | A1          |
| LAP                                                                      | B3                     | B3         | B3          |
| PGM                                                                      | C4                     | C4         | C4          |

What is claimed:

1. A new and distinct variety of strawberry plant, substantially as shown and described.





**FIG.1**



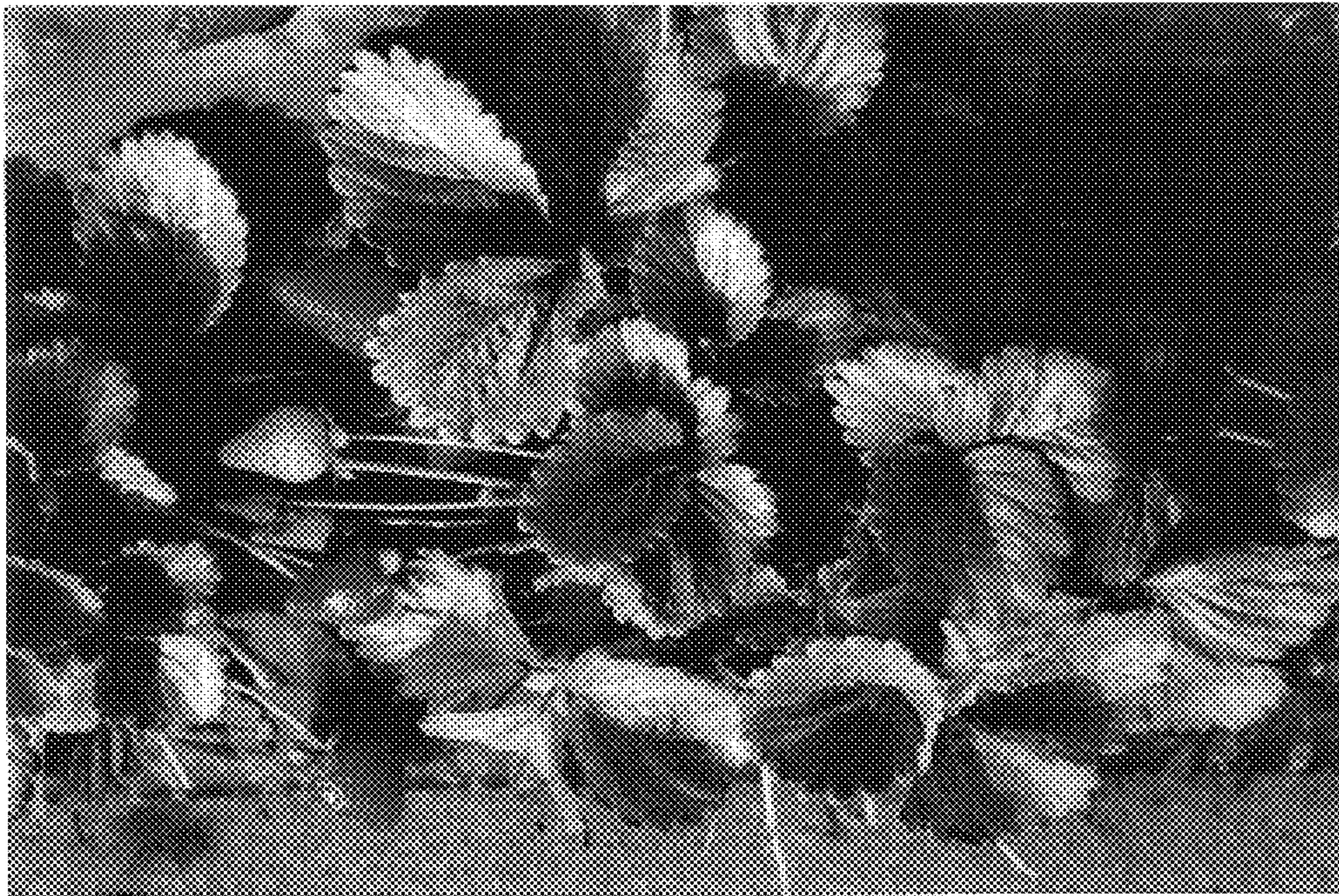


FIG.2



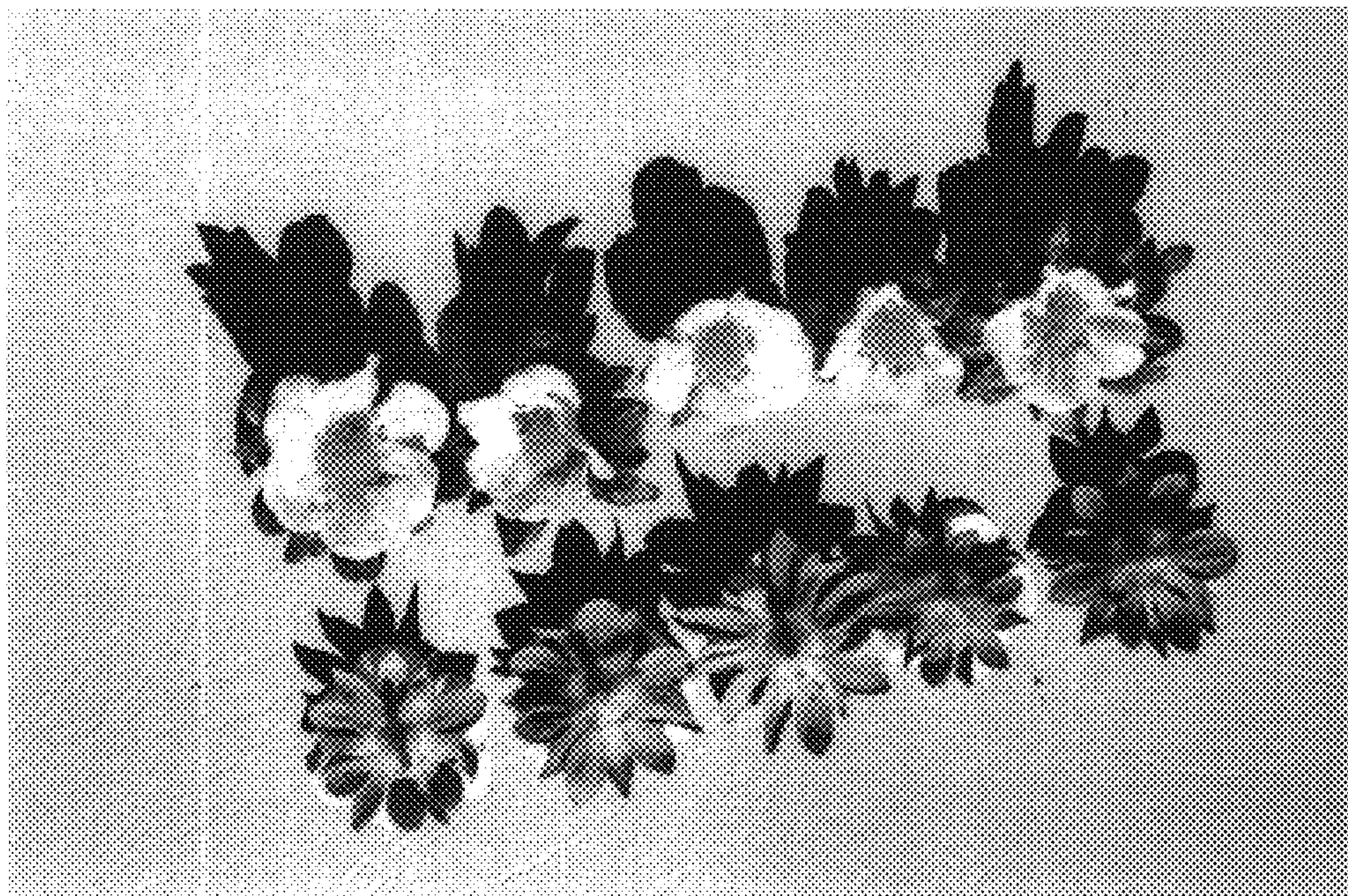


FIG.3



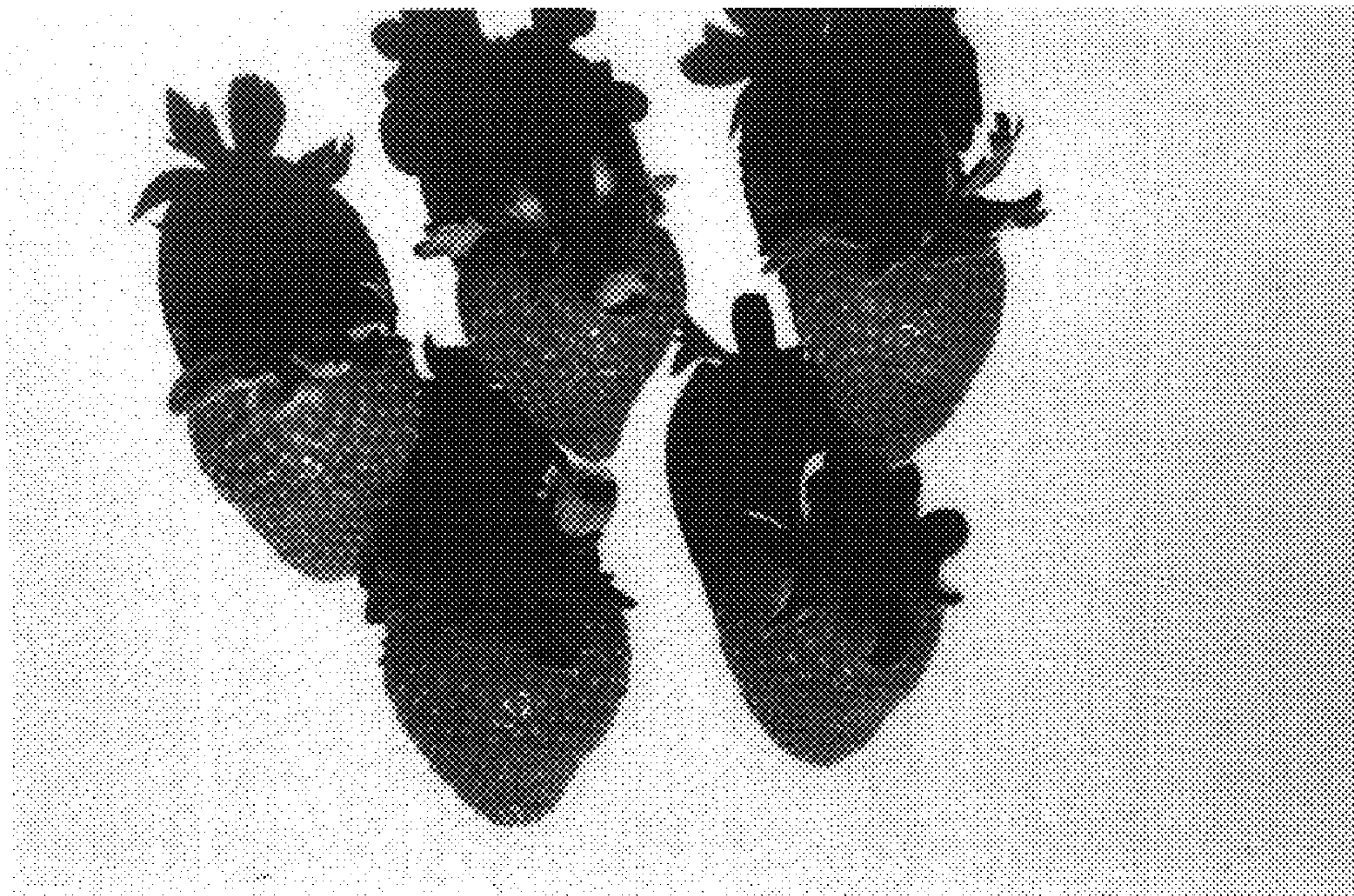


FIG.4



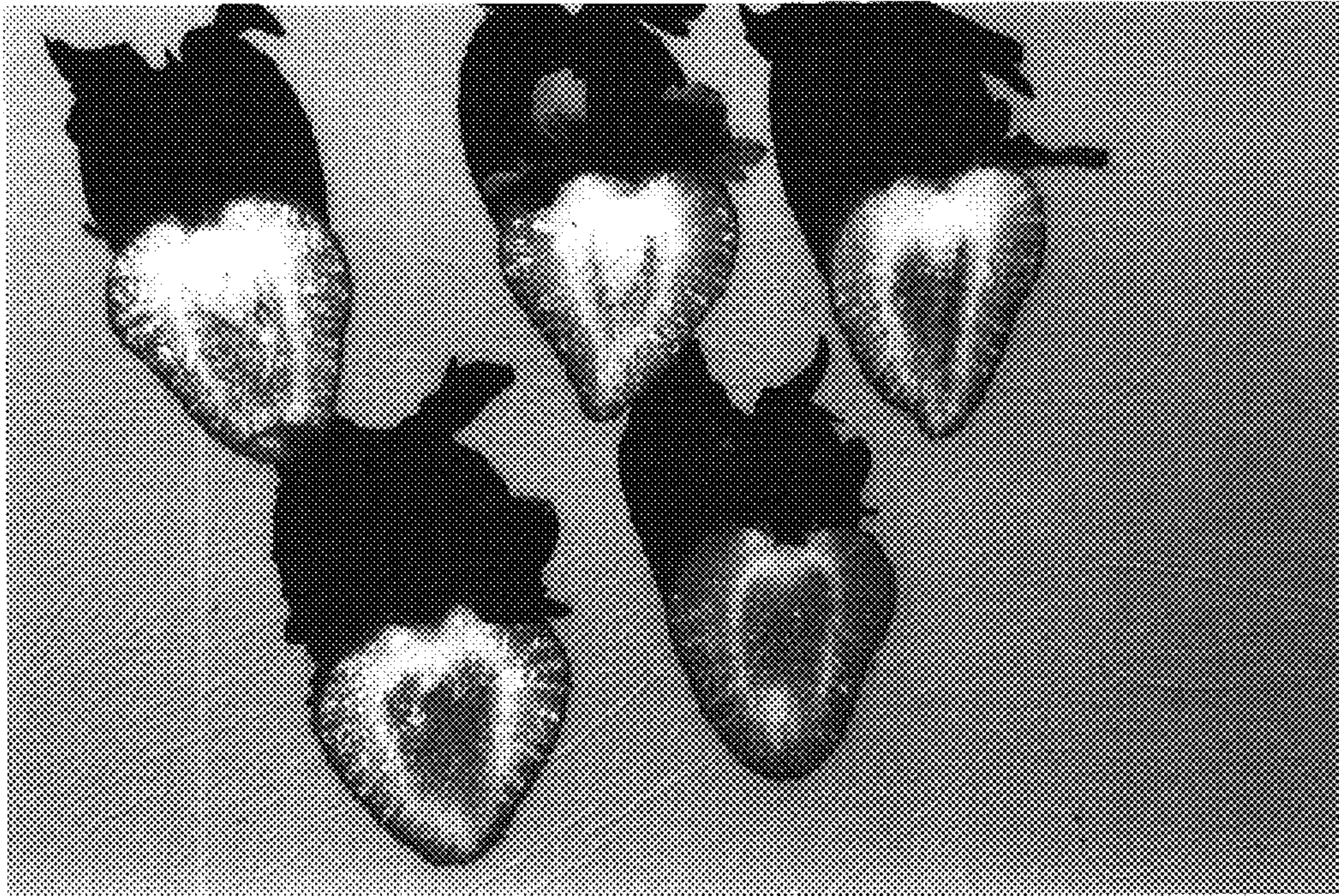


FIG.5